Positive Cosine Powers Disprove the Finite-L^p Form of Conjecture 1

A counterexample to arXiv:1606.09092

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Status. Candidate counterexample, likely valid. It disproves the conjecture exactly as written by exposing the role of the omitted zeroth power in finite L^p; the corrected version with 0 imposed remains open in this packet.

The conjecture

Jaming and Simon [1] define a subset $\Lambda \subset \mathbb{N} = \{0, 1, 2, \ldots\}$ to be a $[-1, 1]$-Müntz–Szász sequence only if $0 \in \Lambda$ and the reciprocal sums over its positive even and odd elements both diverge. Their Conjecture 1 (Conjecture 4.6 in the published numbering) says that, for $p \in [1, \infty]$ and $\theta_1 - \theta_2 \notin \mathbb{Q}$, totality of

$$\{\cos^\lambda 2\pi(t - \theta_1) : \lambda \in \Lambda\} \;\cup\; \{\cos^{\lambda'} 2\pi(t - \theta_2) : \lambda' \in \Lambda'\} \tag{1}$$

in $X_p(0, 1)$ forces both Λ and Λ' to satisfy that definition.

For finite p, a single point carries no L^p information, and the condition $0 \in \Lambda$ is not necessary. This gives a counterexample.

Theorem 1 (counterexample)**.** *Let $1 \le p < \infty$ and let $\theta_1, \theta_2 \in [0, 1)$ satisfy $\theta_1 - \theta_2 \notin \mathbb{Q}$. Set*

$$\Lambda = \Lambda' = \mathbb{N}_+ := \{1, 2, 3, \ldots\}.$$

Then the system in (1) is total in $L^p(0, 1)$, although neither Λ nor Λ' is a $[-1, 1]$-Müntz–Szász sequence under the definition of [1].

Proof. Identify $[0, 1)$ with the circle $\mathbb{T}$. For $\theta \in \mathbb{T}$, let

$$E^p_\theta = \{f \in L^p(\mathbb{T}) : f(\theta + t) = f(\theta - t) \text{ a.e.}\}$$

be the closed reflection-even subspace. Every positive power of $\cos 2\pi(t - \theta)$ belongs to E^p_θ.

The missing constant is nevertheless in their L^p-closed span. Indeed,

$$q_N(t) = 1 - \left(1 - \cos^2 2\pi(t - \theta)\right)^N = \sum_{j=1}^{N} (-1)^{j+1} \binom{N}{j} \cos^{2j} 2\pi(t - \theta). \tag{2}$$

We have $0 \le q_N \le 1$, and $q_N(t) \to 1$ except at the two zeros of the cosine. Dominated convergence gives $q_N \to 1$ in $L^p(\mathbb{T})$.

It follows that the positive powers and all nonnegative powers have the same closed span. That span is exactly E^p_θ: one inclusion is reflection symmetry, while the reverse inclusion follows because the Chebyshev identity

$$\cos 2\pi k(t - \theta) = T_k(\cos 2\pi(t - \theta))$$

puts every reflection-even trigonometric polynomial in the closed span, and such polynomials are dense in E^p_θ. Hence the closed spans of the two families in (1) are respectively $E^p_{\theta_1}$ and $E^p_{\theta_2}$.

We now show that $E^p_{\theta_1} + E^p_{\theta_2}$ is dense. Let $g \in L^{p'}(\mathbb{T})$ annihilate both subspaces, where p' is the conjugate index (with $p' = \infty$ when $p = 1$). Since $g \in L^1(\mathbb{T})$, its Fourier coefficients are defined. Annihilation of $\cos 2\pi k(t - \theta_j) \in E^p_{\theta_j}$ gives, for $j = 1, 2$ and $k \ge 1$,

$$e^{-2\pi i k\theta_j} \widehat{g}(-k) + e^{2\pi i k\theta_j} \widehat{g}(k) = 0. \tag{3}$$

Eliminating $\widehat{g}(-k)$ between the two equations yields

$$\left(e^{4\pi i k\theta_1} - e^{4\pi i k\theta_2}\right) \widehat{g}(k) = 0.$$

The factor in parentheses never vanishes for $k \neq 0$, because $\theta_1 - \theta_2$ is irrational. Thus $\widehat{g}(k) = \widehat{g}(-k) = 0$ for every $k \geq 1$. The constant belongs to both even subspaces, so $\widehat{g}(0) = 0$ as well. Uniqueness of Fourier coefficients in L^1 gives $g = 0$. Hahn–Banach now proves totality of (1).

Finally, $0 \notin \mathbb{N}_+$. Therefore neither chosen exponent set meets the source's definition of a $[-1, 1]$-Müntz–Szász sequence, completing the counterexample. $\square$

Exact scope

The counterexample works for every finite p, including $p = 1$, and its two reciprocal parity sums do diverge; failure of the source condition is solely the missing exponent 0. This is the familiar distinction between uniform and L^p approximation: (2) cannot converge uniformly, because every q_N vanishes where the cosine vanishes. Thus no claim is made against the $C([0, 1])$ endpoint or against the corrected finite-p conjecture in which $0 \in \Lambda \cap \Lambda'$ is imposed explicitly.

Bounded searches through the source title, its exact conjecture, and the two shifted cosine-power terminology found no later paper recording this counterexample. It should nevertheless be viewed as a correction of the statement rather than a resolution of the intended small-divisor problem.

References

[1] P. Jaming and I. Simon, *Müntz–Szász type theorems for the density of the span of powers of functions*, Bull. Sci. Math. 166 (2021), 102933; arXiv:1606.09092.